When Does Symbolic Consensus Help? A Systematic Empirical Study

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Abstract

Recent proposals augment LLM (Large Language Model) reasoning with symbolic methods that extract formal constraints from reasoning traces and use satisfiability solvers for answer selection. We present the first systematic empirical evaluation of symbolic consensus, covering three reasoning domains, five model families, and multiple operating conditions. We instantiate symbolic consensus as SICA (Symbolic Invariant Consensus Aggregation), which extracts first-order logic constraints from K reasoning traces and selects answers via weighted MAX-SAT. Across more than 1,300 problems, SICA provides no significant accuracy gain over standard majority voting (aggregate $|\Delta| \approx 1.5$ percentage points). Extracted constraints are weakly discriminative (mean precision 0.57 vs. 0.33 random baseline for 3-class problems), but this modest signal is insufficient to reliably influence answer selection at practical sample sizes. MAX-SAT optimization is statistically equivalent to simple constraint counting or random weighting. We identify one boundary condition: at temperature T=1.0, SICA achieves +8.3pp over majority voting (p=0.034), but extraction quality degrades from 89% to 76% and output coherence drops to 85%, limiting practical value. We propose C_0, the fraction of problems with multiple candidate answers, as a zero-cost binary diagnostic: C_0<5% guarantees method degeneration to majority voting, while C_0>20% is necessary but insufficient for potential gains.

1 Introduction

Sampling multiple reasoning traces from large language models and aggregating their answers has become standard practice for improving accuracy on complex tasks. Self-Consistency [Wang et al., 2023] selects the majority answer across K sampled chains of thought, providing a simple and effective baseline. Recent work goes further, proposing to extract symbolic constraints from reasoning traces and use formal solvers to aggregate them, potentially exploiting logical structure that majority voting ignores [Pan et al., 2023, Olausson et al., 2023].

The appeal is intuitive: if reasoning traces encode intermediate logical claims, a satisfiability solver should identify the answer most consistent with the full body of extracted knowledge, outperforming a raw head count. Several methods pursue this direction with varying formalisations of constraint extraction and solver integration. Yet a basic practical question remains open: across realistic model families, reasoning domains, and operating conditions, when does symbolic consensus actually outperform majority voting?

No prior work systematically characterises the conditions under which symbolic constraint aggregation helps, degenerates to voting, or actively hurts. Existing evaluations compare symbolic and voting methods on narrow model-domain slices, making it difficult to draw general conclusions about method viability. This gap is consequential for practitioners who must decide whether the substantial overhead of constraint extraction and solving (approximately 49% additional compute in our measurements) is justified by accuracy gains.

We address this gap with a controlled empirical study. We instantiate symbolic consensus as SICA (Symbolic Invariant Consensus Aggregation), which extracts first-order logic (FOL) constraints from K reasoning traces, deduplicates them, and selects answers via weighted MAX-SAT. SICA serves as a concrete, reproducible test bed for the broader question of whether symbolic aggregation adds value over voting. We evaluate SICA against majority voting across three reasoning domains (formal logic, commonsense, science), five model families (four instruction-scale, one full reasoning model), and varied operating conditions (temperature T, sample count K), covering more than 1,300 problems with controlled pairwise comparisons.

Our contributions are as follows:

- The first systematic evaluation of symbolic consensus across multiple models, domains, and operating conditions, establishing that SICA provides no significant gain over majority voting in standard settings (Section 5.1).
- Characterisation of the *capability trap* pattern: instruction-scale models produce ample output diversity (C_0=22–65%), but extracted constraints have only weak discriminative power (precision 0.57), and MAX-SAT over such constraints is equivalent to counting (Sections 5.2–5.3).
- Analysis of boundary conditions showing that extreme temperature (T=1.0) partially breaks the pattern but with degraded extraction quality and output coherence (Section 5.5).
- C_0, a zero-cost binary diagnostic indicating whether symbolic consensus can potentially operate (C_0>20%) or trivially degenerates to voting (C_0<5%) (Section 5.6).

2 Related Work

Self-Consistency and majority voting. Self-Consistency (SC) samples multiple chains of thought and selects the answer receiving the most votes [Wang et al., 2023]. Universal Self-Consistency extends this to free-form answers [Chen et al., 2023]. SC remains the dominant aggregation baseline due to its simplicity and strong empirical performance. Weighted variants assign non-uniform weights to traces based on length or confidence heuristics [Taubenfeld et al., 2024], but evidence for consistent improvement is limited.

Symbolic reasoning for LLMs. Logic-LM translates natural language problems into first-order logic and verifies solutions with theorem provers [Pan et al., 2023]. LINC performs neurosymbolic inference over FOL representations [Olausson et al., 2023]. These methods focus on single-trace symbolic verification rather than multi-trace constraint aggregation. Our work asks whether the multi-trace aggregation dimension adds value beyond what single-trace verification provides.

Process and outcome reward models. Trained verifiers score individual reasoning steps or final answers [Cobbe et al., 2021, Lightman et al., 2023]. Unlike constraint-based aggregation, these methods require supervised training data. They represent a distinct and complementary approach to improving reasoning quality.

Negative results in NLP. Systematic negative-result studies have clarified the limitations of prompting strategies [Reynolds and McDonell, 2021], chain-of-thought elicitation, and various reasoning augmentation techniques. Our work contributes to this tradition by providing controlled evidence on the limits of symbolic consensus.

3 Method: SICA Pipeline

SICA (Symbolic Invariant Consensus Aggregation) aggregates multiple reasoning traces through five stages.

1. Trace generation. Given a problem q, we sample K reasoning traces $\{r_1, \ldots, r_K\}$ from an LLM at temperature T. Each trace contains a chain of thought concluding with a candidate answer.

2. Answer extraction. A deterministic parser extracts the final answer a_i from each trace r_i. Problems where all K traces produce the same answer are classified as *unanimous* and passed directly to majority voting, since no aggregation method can change the outcome.

3. Constraint extraction. For each trace r_i, an LLM-based extractor produces a set of FOL constraints $\mathcal{C}_i = \{c_{i,1}, \ldots, c_{i,m_i}\}$ expressing intermediate logical claims. Each constraint is a first-order formula over the candidate answer space. Extraction success is measured as the fraction of traces yielding at least one parseable constraint.

4. Deduplication. Semantically equivalent constraints across traces are merged, retaining the maximum weight. Deduplication removes redundancy without discarding signal from independent traces that happen to derive the same intermediate claim.

5. MAX-SAT aggregation. We construct a weighted MAX-SAT instance: each unique constraint c_j has weight w_j proportional to extraction confidence. Candidate answers are evaluated by the total weight of constraints they satisfy. The answer maximising satisfied weight is selected. When MAX-SAT produces no discrimination (all candidates satisfy the same constraints), the method falls back to majority vote.

C_0 metric. We define C_0 as the fraction of problems in a dataset where K traces produce more than one distinct candidate answer. C_0 reflects output diversity: when $C_0 \approx 0$, all traces agree and any aggregation method reduces to the unanimous answer. C_0 is computable with zero overhead from the answer extraction stage and requires no constraint extraction.

4 Experimental Setup

Models. We evaluate seven models spanning four categories. *Qwen family (instruction-scale)*: Qwen2.5-7B-Instruct, Qwen2.5-14B-Instruct, Qwen2.5-72B-Instruct-AWQ (4-bit quantised). *Other instruction-scale*: Mistral-7B-Instruct-v0.3, LLaMA-3.1-8B-Instruct. *Distilled reasoning*: DeepSeek-R1-Distill-Qwen-7B, which we initially expected to behave as a reasoning model but reclassified as instruction-scale after observing C_0=58% (far above the $< 5\%$ expected for reasoning models). *Full reasoning*: QwQ-32B.

Domains. ProofWriter (deductive reasoning over formal rules), FOLIO (first-order logic NLI), StrategyQA (commonsense binary reasoning), and ARC-Challenge (science multiple-choice). The primary evaluation set combines ProofWriter and FOLIO ("D012") for formal logic coverage.

Table 1: Main results: SICA vs. Self-Consistency (SC) across models and domains. C_0: fraction of problems with multiple candidate answers. Δ: SICA − SC in percentage points. p: McNemar's exact test. Best per-row result in **bold**.

Model	Domain	N	C_0%	SC	SICA	Δpp	p
Qwen-14B	PW+FOLIO	804	38	**76.2**	76.5	+0.3	n.s.
Qwen-14B	StrategyQA	60	10	53.3	**56.7**	+3.3	0.16
Qwen-14B	ARC-Chal.	60	5	**96.7**	95.0	−1.7	—
Qwen-7B	PW+FOLIO	60	65	56.7	**60.0**	+3.3	0.41
Qwen-72B	PW+FOLIO	60	22	85.0	**86.7**	+1.7	0.65
Mistral-7B	PW+FOLIO	60	50	48.3	48.3	0.0	1.00
LLaMA-8B	PW+FOLIO	60	62	60.0	**68.3**	+8.3	0.096
DeepSeek-R1-7B	PW+FOLIO	60	58	**71.7**	68.3	−3.3	0.48
QwQ-32B	Mixed	46	2.6	≡ SC		0.0	—

Default configuration. K=12 traces per problem, temperature T=0.7, unless stated otherwise. Temperature and K are varied in ablation experiments.

Statistical testing. We use McNemar's exact test for paired comparisons on the same problem set. For experiments involving multiple comparisons across model families, we apply Bonferroni correction (adjusted $\alpha \approx 0.01$ for five families).

5 Results

5.1 Main Results

Table 1 presents SICA versus SC accuracy across all model-domain configurations.

Across all standard conditions, SICA accuracy falls within a few percentage points of SC. The largest positive delta (+8.3pp, LLaMA-8B) does not reach significance after Bonferroni correction for multiple model comparisons (McNemar p=0.096, corrected threshold $\approx$ 0.01). This result is further confounded by 48 truncated traces out of 720 total (6.7% truncation rate), which may artificially inflate diversity. The largest negative delta (−3.3pp, DeepSeek-R1-7B) occurs with the highest truncation rate in the study (72/720 = 10.0%). No model-domain combination shows SICA significantly outperforming SC at α=0.05 after correction.

QwQ-32B represents the converged regime. With effective C_0=2.6%, nearly all problems produce unanimous traces, and SICA trivially reduces to SC. Full reasoning models, by design, produce deterministic-like chains that eliminate the diversity symbolic methods require.

Within the Qwen family, C_0 decreases monotonically with model scale (7B: 65%, 14B: 38%, 72B: 22%, QwQ-32B: 2.6%), but the SICA-SC gap does not follow a corresponding trend (+3.3, +0.3, +1.7, 0.0pp). Diversity reduction does not predict method performance.

DeepSeek-R1-Distill-Qwen-7B was initially expected to function as a reasoning model (C_0<5%). In practice, at 7B parameter capacity, distillation does not produce the deterministic reasoning chains characteristic of full reasoning models. C_0=58% places it squarely in the instruction-scale cluster, and we reclassify it accordingly.

Table 2: Constraint precision on multi-candidate problems (Qwen-14B, PW+FOLIO, n=307). Precision measures the fraction of extracted constraints that correctly discriminate the ground-truth answer from alternatives. Random baseline: 0.33 for 3-class, 0.50 for binary.

Domain	N	Mean	Std	Median
ProofWriter	—	0.55	—	—
FOLIO	—	0.62	—	—
Overall	307	0.57	0.23	0.58

Table 3: Constraint ablation (Qwen-14B, PW+FOLIO, n=804, T=0.7). MAX-SAT: full SICA pipeline. Count-only: constraints weighted equally, no satisfiability reasoning. Random-weight: constraints with random weights. SC: standard majority voting.

Method	Accuracy (%)	Δ vs. SC (pp)	p vs. SC
MAX-SAT (SICA)	76.5	+0.3	—
Count-only	76.4	+0.2	—
Random-weight	76.3	+0.1	—
SC (baseline)	76.2	—	—
MAX-SAT vs. Count-only: p=0.85; agreement on 802/804 problems. MAX-SAT vs. Random: p=0.72.			

5.2 Constraint Quality Analysis

Table 2 characterises per-problem constraint precision across the 307 multi-candidate problems from the main Qwen-14B experiment (n=804).

Mean constraint precision is 0.57 (std 0.23), significantly above the 0.33 random baseline for 3-class problems. Constraints are *weakly discriminative*, not random. However, precision shows no correlation with vote entropy (R^2=0.008, p=0.12), meaning constraint quality does not improve when output diversity increases. FOLIO constraints (precision 0.62) are somewhat more accurate than ProofWriter constraints (0.55), consistent with FOLIO's more explicit logical structure.

A precision of 0.57 explains the aggregate null result quantitatively. At K=12 with $C_0 \approx 38\%$, approximately 4 multi-candidate problems per 60-question set have constraints that could influence the outcome. With constraints correct 57% of the time, the expected number of correct answer flips per experiment is approximately 2, which is indistinguishable from sampling noise at these sample sizes.

5.3 Mechanism Ablation

Table 3 compares four aggregation strategies on the main experiment data (n=804) to isolate the contribution of MAX-SAT reasoning.

MAX-SAT (76.5%) is statistically indistinguishable from count-only aggregation (76.4%), random-weight aggregation (76.3%), and standard majority voting (76.2%). The four methods disagree on only 2 of 804 problems. The logical structure encoded in constraints is not exploited by the solver in any meaningful way: MAX-SAT degenerates to a constraint counter, and constraints provide no incremental signal beyond raw vote counts.

Table 4: Tiebreak accuracy on tied questions (Qwen-14B, K=12, n_{tied}=5 of 60). Tie fraction: 8.3%.

Tiebreak Method	Accuracy on Ties
SICA (constraint-based)	80% (4/5)
Alphabetical	60% (3/5)
Random (expected)	50%
First-seen	40% (2/5)

Table 5: Temperature ablation (Qwen-14B, D012, n=60, K=12). Default operating condition is T=0.7. Extraction rate: fraction of traces yielding parseable constraints. Coherence: fraction of traces producing coherent, complete outputs.

T	C_0%	SC	SICA	Δpp	p	Extr.	Coher.
0.3	18	73.3	75.0	+1.7	0.65	93%	∼98%
0.5	30	73.3	75.0	+1.7	0.55	91%	∼98%
0.7	38	76.2	76.5	+0.3	n.s.	89%	∼98%
1.0	72	73.3	81.7	+8.3	**0.034**	76%	85%

5.4 Tiebreak Analysis

Table 4 examines SICA's performance specifically on tied questions, where majority voting must break ties arbitrarily.

On the small subset of tied questions (5 of 60, tie fraction 8.3%), SICA selects the correct answer 80% of the time versus 50% for random tiebreaking. Constraint-based tiebreaking has genuine value on individual ties. However, ties are too rare to materially affect overall accuracy. The effective contribution is $0.80 \times 0.083 \approx 6.6\%$ of possible gain, translating to at most ∼2.5pp under ideal conditions. At even K (where ties can occur), the tie rate varies from 5.0% (K=20) to 26.7% (K=12), but even at peak tie rate the overall accuracy impact remains within noise.

5.5 Boundary Conditions: Temperature

Table 5 presents the temperature ablation on Qwen-14B over 60 D012 problems. This is the critical experiment for understanding when symbolic consensus can differ from voting.

Temperature T=1.0 produces the only statistically significant SICA advantage in the entire study: +8.3pp (McNemar p=0.034) with a 7:1 discordant ratio favouring SICA. C_0 increases from 38% at T=0.7 to 72% at T=1.0, confirming the strong link between temperature and diversity (r=0.98). The gap is monotonically increasing with temperature.

This gain comes with substantial costs. Extraction rate drops from 89% (T=0.7) to 76% (T=1.0): nearly one quarter of constraint extraction attempts produce empty or unparseable output. Answer coherence drops to 85%, meaning 15% of generated traces are partially incoherent. When extraction retries are factored in, the estimated overhead at T=1.0 rises to approximately 65% (vs. 49% at default temperature), increasing the per-question compute cost by roughly 30%. The T=0.7 reference row reports the main experiment (n=804) for context; all other rows use n=60.

The practical implication is that the boundary condition exists in theory but is operationally costly. A practitioner seeking the marginal +8.3pp would need to tolerate incoherent outputs,

Table 6: C_0 clusters and corresponding SICA-SC gaps. Low-C_0 cluster: reasoning models and high-confidence domains. High-C_0 cluster: instruction-scale models on challenging domains.

Cluster	C_0 range	Mean gap (pp)	N points
Low C_0	0–10%	−0.9	3
High C_0	22–72%	+1.8	9
Linear regression: R^2=0.08, slope= 0.09pp per %C_0, p=0.37.			

higher extraction failure rates, and additional compute. For comparison, simply running more traces at the default temperature (K=16 or K=20) achieves similar variance reduction without these quality degradations.

Lower temperatures (T=0.3, T=0.5) produce small positive deltas (+1.7pp each) that are not significant (p>0.5), consistent with the default-temperature null result.

5.6 C_0 as Operating Regime Indicator

Across all 12 model-domain-temperature conditions, C_0 values cluster into two distinct regimes (Table 6).

The low-C_0 cluster (C_0=0–10%) comprises QwQ-32B (C_0=2.6%), ARC-Challenge (C_0=5%), and StrategyQA (C_0=10%). Mean gap in this cluster is −0.9pp: SICA trivially reduces to voting. The high-C_0 cluster (C_0=22–72%) comprises instruction-scale models on formal logic domains plus the temperature extreme. Mean gap is +1.8pp but with enormous variance (from −3.3 to +8.3pp), and no individual comparison is significant except T=1.0.

Linear regression of gap on C_0 yields R^2=0.08 (p=0.37): C_0 provides no gradient-level predictive power for the SICA-SC gap. The 10–22% range contains only one data point (C_0=18% at T=0.3), leaving a gap between clusters.

C_0 therefore functions as a binary indicator. C_0<5% reliably predicts SICA ≡ SC (method degeneration). C_0>20% indicates that symbolic consensus can potentially differ from voting, but does not predict whether the difference will be positive, negative, or zero. This makes C_0 useful as a zero-cost go/no-go check before investing in constraint extraction, but not as a quantitative performance predictor.

6 Discussion

The capability trap as descriptive pattern. Across our experiments, we observe a recurring pattern that we label the *capability trap*: conditions that enable symbolic consensus to differ from majority voting (high output diversity, C_0>20%) do not ensure that the difference is beneficial. Instruction-scale models produce abundant diversity (C_0=22–65%), but extracted constraints at precision 0.57 are insufficient to reliably steer answer selection. We emphasise that this is a descriptive label for an empirical observation, not a mechanistic explanation. The decoupling of diversity and constraint quality may stem from the extraction process, the FOL formalism, or inherent limitations of current-generation LLM reasoning.

Practical guidelines. For practitioners considering symbolic aggregation, our results suggest a simple decision framework. First, compute C_0 from raw answer extractions at zero cost. If C_0<5% (strong reasoning models, easy tasks), symbolic methods will trivially degenerate to voting. If C_0>20% under standard temperatures, symbolic methods may differ from voting but are unlikely to improve accuracy, while incurring approximately 49% additional compute (Appendix B).

The T=1.0 exception and its limits. The temperature ablation reveals that extreme diversity (T=1.0, C_0=72%) can produce a significant SICA advantage. This finding prevents us from claiming that the capability trap is absolute. However, the associated quality degradation (extraction rate 76%, coherence 85%) and the fact that this result comes from a single n=60 experiment temper its practical significance. Replication across models and larger sample sizes is needed before recommending T=1.0 as an operating point.

Alternative aggregation directions. The Logic-LM comparison (Appendix D) shows that Logic-LM with K repetitions and majority voting achieves 80.9% on FOLIO vs. SICA's 79.4%. Single-trace FOL verification repeated K times may be more effective than multi-trace constraint aggregation. Inverse-length weighted voting (Appendix C) matches SICA at zero extraction cost (77.0% vs. 76.5%). These results suggest that lightweight quality proxies, or per-trace verification, may be more productive directions than cross-trace constraint consensus.

Limitations. (1) We evaluate one instantiation of symbolic consensus (FOL constraints, MAX-SAT solver). Other constraint languages or aggregation strategies might yield different results. (2) The largest powered test (n=804) covers only Qwen-14B on ProofWriter and FOLIO. Cross-model tests at $n\geq$200 exist only for LLaMA-8B. (3) We test only one full reasoning model (QwQ-32B); additional reasoning models (o1-series, Gemini thinking models) would strengthen the converged-regime claim. (4) The T=1.0 boundary condition has been tested only on Qwen-14B with n=60. (5) The constraint ablation (Table 3) was performed at T=0.7; results may differ at extreme temperatures where the constraint quality distribution changes.

7 Conclusion

We present the first systematic empirical study of symbolic consensus for LLM reasoning aggregation. Across three domains, five model families, and over 1,300 problems, symbolic constraint aggregation via SICA provides no significant accuracy improvement over majority voting in standard operating conditions. Extracted constraints are weakly discriminative (precision 0.57) but insufficient to reliably influence answer selection. The MAX-SAT solver degenerates to a constraint counter: it is statistically equivalent to counting or random weighting. At extreme temperatures (T=1.0), a marginal gain appears (+8.3pp, p=0.034) but is accompanied by degraded extraction quality and output coherence. We contribute C_0 as a practical, zero-cost binary diagnostic for practitioners deciding whether to invest in symbolic aggregation. Our findings indicate that, for the current generation of LLM constraint extraction, simple majority voting remains the method of choice.

A K Sensitivity Analysis

Table 7 presents the effect of varying K on SICA vs. SC (Qwen-14B, D012, n=60).

Table 7: K ablation (Qwen-14B, D012, n=60, T=0.7). Tiebreak rate: fraction of problems producing exact ties at this K.

K	SICA	SC	Δpp	Tiebreak Rate
6	75.0	73.3	+1.7	18.3%
8	75.0	75.0	0.0	—
11	75.0	75.0	0.0	—
12	73.3	75.0	−1.7	26.7%
16	71.7	73.3	−1.7	—
20	71.7	73.3	−1.7	5.0%

At odd K=11, where exact ties cannot occur, the gap is precisely zero. This confirms that SICA's potential value comes entirely from tiebreaking. At K=6 (small sample), the +1.7pp positive delta falls within the bootstrap 95% CI $[-5.0, +8.3]$pp, compatible with zero effect. At $K \geq 12$, SICA shows a consistent −1.7pp deficit, likely from noise injection by low-quality constraints in larger trace sets. The pattern is non-monotonic (positive at low K, zero at odd K, negative at high K), making a blanket "SICA $\leq$ SC at all K" statement inaccurate. The correct characterisation is *K-dependent with no consistent advantage*.

B Overhead Analysis

Table 8 breaks down wall-clock time per question from the main experiment (n=804).

Table 8: Per-question wall-time breakdown (Qwen-14B, n=804, T=0.7).

Component	Mean time (s)	% of total
Trace generation	31.6	51.3%
Constraint extraction	29.0	47.0%
Deduplication	1.0	1.6%
MAX-SAT solving	0.003	<0.01%
Total	61.6	100%

SICA's overhead relative to SC (which requires only trace generation) is 48.7%, dominated by the constraint extraction LLM call. MAX-SAT solving time is negligible. The overhead-to-gain ratio at default temperature is 48.7%/0.3pp $\approx$ 162: each percentage point of (non-significant) gain costs 162% in additional compute. Median overhead is 48.9%, with the 90th percentile at 57.6%. ProofWriter questions incur 51.3% overhead on average vs. 38.6% for FOLIO, reflecting longer constraint extraction prompts for deductive rules.

C Weighted SC Baselines

Table 9 compares SICA against simple weighted voting heuristics (n=804).

Table 9: Weighted SC variants (Qwen-14B, PW+FOLIO, n=804, T=0.7).

Method	Accuracy (%)	Δ vs. SC (pp)	p vs. SC
Inverse-length	77.0	+0.8	0.42
SICA (MAX-SAT)	76.5	+0.3	—
SC (uniform)	76.2	—	—
Length-weighted	75.3	−0.9	0.16
Constraint-count	75.3	−0.9	—

Inverse-length weighting (favouring shorter, more confident traces) achieves 77.0%, slightly above SICA's 76.5%. All differences are non-significant. A zero-cost heuristic based on trace length matches or exceeds the full SICA pipeline, reinforcing the conclusion that constraint extraction does not add value at default settings.

D Logic-LM Comparison

Table 10 compares SICA against a Logic-LM baseline on FOLIO (n=204).

Table 10: Logic-LM × K vs. SICA vs. SC on FOLIO (n=204, K=12).

Method	Accuracy (%)	p vs. SICA
Logic-LM × K + vote	80.9	0.38
SICA (MAX-SAT)	79.4	—
SC (majority vote)	79.4	—
Single-trace Logic-LM	77.5	—

Logic-LM repeated K times with majority voting achieves 80.9% vs. SICA's 79.4%. The difference is not significant (p=0.38), but the directional advantage of per-trace verification over multi-trace constraint aggregation is notable. Logic-LM performs a complete FOL consistency check per trace (pass/fail), which has stronger discriminative power than SICA's soft constraint aggregation: an internally inconsistent trace is excluded entirely by Logic-LM, while SICA still incorporates its constraints.

E GSM8K Pilot

A 30-question pilot on GSM8K (Qwen-14B, K=12) yields: SICA 80.0% vs. SC 76.7% ($\Delta = +3.3$pp, p=0.56, McNemar). C_0=27% (moderate), multi-candidate rate 23%, extraction rate 85%. At n=30, the standard error is approximately 8pp, making this result uninformative. GSM8K is included as a pilot data point for domain coverage; it does not alter any main-text claims.

F Constraint Case Studies

We qualitatively analyse 8 representative questions from the main experiment. Of these, 3 produce semantically meaningful constraints, 2 produce correctly discriminative constraints that steer SICA

to the right answer, 3 produce non-discriminative constraints (tautological or irrelevant), and 2 produce actively misleading constraints that steer SICA away from the correct answer.

Constraint categories: 3 tautological, 2 partially correct, 2 fully correct, 1 contradicting ground truth. The two fully correct cases both come from FOLIO problems with explicit formal logical structure, where FOL extraction naturally aligns with the problem's own representation. The misleading case involves a ProofWriter problem where the extracted constraint encodes a premise not present in the original rules.

These case studies illustrate that constraints *can* be discriminative on individual problems (consistent with the aggregate precision of 0.57), but the mixture of helpful, neutral, and harmful constraints yields no net improvement when aggregated across a full problem set.

G C_0 Diagnostic Full Data

Table 11 lists all 12 data points used for the C_0 analysis in Section 5.6.

Table 11: All (model/condition, domain, C_0, SICA-SC gap) data points.

Condition	Domain	C_0 (%)	Gap (pp)
QwQ-32B	Mixed	2.6	0.0
Qwen-14B	ARC	5.0	−1.7
Qwen-14B	StrategyQA	10.0	+3.3
Qwen-14B T=0.3	PW+FOLIO	18.0	+1.7
Qwen-72B	PW+FOLIO	22.0	+1.7
Qwen-14B GSM8K	GSM8K	27.0	+3.3
Qwen-14B T=0.7	PW+FOLIO	38.0	+0.3
Mistral-7B	PW+FOLIO	50.0	0.0
DeepSeek-R1-7B	PW+FOLIO	58.0	−3.3
LLaMA-8B	PW+FOLIO	62.0	+8.3
Qwen-7B	PW+FOLIO	65.0	+3.3
Qwen-14B T=1.0	PW+FOLIO	72.0	+8.3

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